

Dynamic Pricing & Revenue Management SaaS

Product Requirements Document (PRD)

Prepared For	Client
Project Budget	USD 5,000
Estimated Duration	10 Days
Payment Structure	5 Fixed-Price Milestones

1. Project Overview

Dynamic Pricing & Revenue Management SaaS is a multi-tenant platform that helps businesses manage and optimize pricing based on demand, availability, occupancy, seasonality, and custom pricing rules.

The platform provides a centralized dashboard where businesses can manage products/services, pricing rules, availability, demand data, price changes, and revenue performance.

The system can automatically calculate or recommend suitable prices based on configured business rules and market conditions, while allowing managers to manually review and override prices when required.

The platform is designed for businesses such as hotels, vacation rentals, event venues, parking operators, rental businesses, and other businesses where prices change based on demand and availability.

2. Objectives

- Provide a centralized system for managing dynamic prices.
- Allow businesses to create custom pricing rules.
- Adjust prices based on demand and availability.
- Support seasonal and date-based pricing.
- Provide revenue and pricing analytics.
- Allow managers to manually approve or override recommended prices.
- Identify high-demand and low-demand periods.
- Provide AI-assisted pricing recommendations.
- Support multiple organizations with isolated data.
- Reduce manual pricing decisions and improve revenue visibility.

3. User Roles

Admin

- Manage organization settings.
- Manage users and permissions.
- Configure pricing settings.
- Manage products/services.
- View all analytics.

Revenue Manager / Operations Manager

- Create pricing rules.

- Monitor demand and availability.
- Review recommended prices.
- Approve or override price changes.
- View revenue analytics.

Staff / Manager

- View assigned products/services.
- Update availability.
- View current pricing.
- Manage approved price changes.

Viewer / Client

- Read-only access to pricing dashboards and reports.

4. Functional Requirements

Authentication & Organization Management

- Email/password signup and login.
- Role-based access control.
- Multi-tenant organization setup.
- Organization-specific users, products, pricing and analytics.
- Password reset and account management.

Product / Inventory Management

Businesses can create and manage items that require dynamic pricing.

- Name
- Category
- Description
- Base price
- Availability
- Capacity
- Location
- Status
- Minimum/maximum price
- Pricing configuration

Deluxe Room
 Base Price: \$100
 Minimum Price: \$70
 Maximum Price: \$180
 Availability: 5/20

Availability & Occupancy

- Available units
- Occupied units
- Total capacity
- Occupancy percentage

- Date/time availability
- Booking or reservation volume

Total Rooms: 100
 Occupied: 85
 Occupancy: 85%

Pricing Rules Engine

Managers can create custom pricing rules.

If occupancy > 80% → Increase price by 15%
 If occupancy < 30% → Decrease price by 10%
 Weekend → Increase price by 20%
 Peak Season → Apply seasonal pricing

- Conditions
- Actions
- Priority
- Start/end date
- Active/inactive status
- Minimum/maximum price limits

Seasonal Pricing

- Peak season
- Off-season
- Holidays
- Weekends
- Special events
- Custom date ranges

The system automatically applies the relevant pricing rules.

Dynamic Price Calculation

- Base price
- Demand
- Availability
- Occupancy
- Season
- Day of week
- Custom rules

The pricing engine combines these inputs and generates a calculated/recommended price.

Base Price \$100
 High Occupancy +15%
 Weekend +10%
 Recommended Price = \$125

The final price should always respect configured minimum and maximum price limits.

5. Pricing Dashboard

The main dashboard will provide an operational view of pricing and revenue.

KPIs

- Current Revenue
- Average Price
- Occupancy
- Revenue Growth
- Price Changes
- High-Demand Items
- Low-Demand Items

Charts

- Revenue trends
- Occupancy trends
- Average price trends
- Demand trends
- Price change history

6. Price Calendar

A calendar-based interface will allow managers to see pricing across dates.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Room A	\$100	\$100	\$110	\$110	\$130	\$150	\$140
Room B	\$90	\$90	\$100	\$100	\$120	\$140	\$130

- View prices by date.
- Compare prices.
- Identify high-demand dates.
- Apply manual overrides.
- Review upcoming pricing changes.

7. Price Approval & Override

Pricing Recommendation
↓
Manager Review
↓
Approve / Modify / Reject
↓
Price Published

- User
- Previous price
- New price
- Reason
- Date/time

Manual overrides should be recorded for auditability.

8. Revenue Analytics

- Revenue by product
- Revenue by date

- Revenue by season
- Average selling price
- Occupancy rate
- Demand trends
- Price performance
- Revenue impact of price changes

Reports should support filtering by product, category, date range, location, and season.

9. Notifications

- High-demand periods
- Low occupancy
- Price rule conflicts
- Prices reaching maximum/minimum limits
- Pending price approvals
- Important revenue anomalies

Notifications can initially support in-app and email.

10. AI & Automation

AI should be used as an assistant to the pricing workflow rather than as an independent chatbot.

AI Price Recommendations

- Historical demand
- Occupancy
- Previous pricing
- Seasonal patterns
- Availability

Demand Forecasting

- Expected high-demand periods
- Expected low-demand periods
- Potential revenue opportunities

Pricing Anomaly Detection

The system can flag unusual situations, such as occupancy increasing rapidly while the current price has not changed.

Revenue Insights

AI can generate short explanations such as: "Weekend demand is higher than the previous four weeks. Increasing the price within the configured range may improve revenue."

All AI recommendations remain subject to configured pricing limits and manager approval.

11. Frontend Design

The application will use a clean, data-focused SaaS dashboard similar in principle to an operations-console approach, prioritizing quick status scanning, reusable components, dashboards, charts, and responsive layouts.

Design System

- React

- Tailwind CSS
- Reusable cards
- Tables
- Charts
- Modals
- Status badges
- Forms

Main Screens

- Login/Register
- Dashboard
- Products/Inventory
- Pricing Rules
- Price Calendar
- Price Recommendations
- Analytics
- Notifications
- Users & Roles
- Settings

Responsive Design

- Desktop
- Tablet
- Mobile

12. Non-Functional Requirements

- Secure authentication.
- Role-based API access.
- Tenant-level data isolation.
- Encryption in transit.
- Audit logging for pricing changes.
- Responsive UI.
- Optimized dashboard performance.
- Scalable architecture for multiple organizations.
- Reliable pricing calculation.
- Validation to prevent prices outside configured limits.

13. Payment Integration

Payment processing is not required for this engagement. The architecture should remain flexible enough to add subscription billing for organizations in a future phase without major architectural changes.

14. Project Budget

Milestone	Amount
Milestone 1	\$500
Milestone 2	\$1,000
Milestone 3	\$1,200
Milestone 4	\$1,300
Milestone 5	\$1,000
Total	\$5,000

15. Project Milestones

Milestone 1 — Foundation & Authentication

Day 1 — \$500

- Project architecture
- Repository setup
- Frontend/backend setup
- Database setup
- Authentication
- Role-based access
- Multi-tenant organization structure
- Initial dashboard shell
- Navigation
- Base UI components
- Staging deployment

Client receives: Working staging link with login, roles and basic dashboard structure.

Milestone 2 — Products, Availability & Pricing Rules

Days 2–3 — \$1,000

- Product/inventory management
- Product categories
- Base pricing
- Minimum/maximum price
- Availability management
- Occupancy calculation
- Seasonal pricing
- Custom pricing rules
- Pricing rules management UI

Client receives: Working staging system where products can be created, availability can be configured and pricing rules can be created.

Milestone 3 — Dynamic Pricing Engine & Price Calendar

Days 4–5 — \$1,200

- Dynamic pricing calculation engine
- Demand-based pricing
- Occupancy-based pricing
- Seasonal pricing
- Rule priority handling
- Price calendar
- Price recommendations
- Manual price override
- Price history
- Approval workflow

Client receives: A complete pricing workflow where rules generate recommended prices and managers can review, modify and approve them.

Milestone 4 — Analytics, AI & Notifications

Days 6–8 — \$1,300

- Revenue dashboard
- Occupancy analytics
- Demand analytics
- Average price analytics
- Revenue trends
- AI price recommendations
- Demand forecasting
- Pricing anomaly detection
- AI-generated revenue insights
- In-app notifications
- Email notifications
- Audit logging

Client receives: Complete analytics dashboard with AI-assisted pricing and revenue insights.

Milestone 5 — Testing, Reports & Go-Live

Days 9–10 — \$1,000

- CSV/PDF report export
- Security review
- Performance optimization
- Responsive UI polish
- Full workflow testing
- Bug fixing
- Production deployment
- Domain configuration
- SSL
- Production database

- Environment variables
- Final smoke testing
- Documentation
- Knowledge-transfer session

Client receives: Live production application, source code, documentation and handover session.

16. Acceptance Criteria

- All five milestones are implemented.
- Authentication and roles work correctly.
- Organizations have isolated data.
- Products and availability can be managed.
- Pricing rules can be created and applied.
- Dynamic prices are calculated correctly.
- Managers can approve/override prices.
- Analytics display pricing/revenue data.
- AI recommendations work within configured limits.
- Notifications work.
- Reports can be exported.
- Application is responsive.
- Core workflows are tested.
- Application is deployed to production.
- Source code and environment documentation are delivered.

17. Deployment & Go-Live

- Production hosting setup
- Domain connection
- DNS configuration
- SSL
- Production database
- Database security
- Backup configuration
- Environment variables/secrets
- Production deployment
- Final smoke testing
- Go-live verification

18. How We Will Proceed

```

Build
↓
Deploy to Staging
↓
Client Reviews
↓

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Feedback
↓
Revision
↓
Milestone Approval
↓
Next Milestone

A working staging link will be provided after each milestone rather than only providing progress updates.

Minor revisions within the agreed milestone scope are included. New features or scope changes will be treated separately.

19. Future Scope

- Direct integration with booking/reservation platforms
- External market-price data integration
- Competitor price monitoring
- Advanced demand forecasting
- Automated price publishing
- Mobile application
- Subscription/billing system
- Advanced revenue optimization
- Multi-currency support
- Advanced AI pricing models